Re: preliminary VALOR TN

Mark Hartz [mark.hartz@ipmu.jp]
Inviato: venerdì 8 aprile 2016 22.06
A: cgiganti@lpnhe.in2p3.fr
Cc: Davide Sgalaberna; Raj.Shah@physics.ox.ac.uk

Hi Davide, Raj, Claudio,

My comments are below. In general, I think the note looks good, but please review the nue-bar appearance and numu-bar disappearance sections. In some parts, the explanation is a little bit difficult to follow. We may benefit from leaving them out for now and including them after we have a chance to polish the text and make comparisons between analyses.

Best,
Mark

43: Add a note that the FC method will be shown in a later version of the note. Done
103: Add (FHC) and (RHC) in parenthesis since they are used later Done
113: Reference 29 is technical note 230 Done
136: contest -> context Done
385: The description of the flux parameters is a bit confusing. It would be better to specify the horn polarity or say (anti)neutrino enhanced beam modes. Done
392: You may consider dropping the description of the Cholesky method. The important thing to note is that parameters are thrown according to their multivariate normal distribution, and that correlations are accounted for. You can just state that the Cholesky method is used to do this. Done
401: change to "...neutrino energy bins' normalizations." Done
431: Specify in text that it was the analysis presented at the 2015 summer conferences Done
433-437: Can you list the numerical impact of each item on the RHC 1Re prediction? Done
468: The table reference is broken Done
482: I think what is considered a oscillation "nuisance" parameter depends on which contour is being drawn. Can you make it clear that all parameters not plotted in a given contour are nuisance parameters and are marginalized? Done
520: Toys MC repeats Done
Fig 6: It is not urgent, but on such plots it is better to specify that it is after the ND280 fit so it is not confused with the SK fit TODO
Fig 8: The last sentence in the caption is critical for understanding these plots. I suggest putting that information near the beginning of the caption. Done
Tables 17-20: It is not urgent, but it would be nice to combine these into a single table or two tables so the systematic errors can easily be compared between samples. TODO
Table 21: Make it clear that these are fractional errors on the ratio. Actually since previous tables were percentages, it would be better to show them as percentages Done
534: We show the delta-chi^2 between the normal and inverted hierarchy cases, so I don't thing we are just measuring the absolute value of delta-m^2 Done
541: You may use theta_l for the lepton angle throughout the document to differentiate from the fitted oscillation parameters Done
Fig 28: Show the contours with the global minimum Done
774: This introduction is a little bit unclear. I think you should make it clear that you use the same analysis framework and samples used in the joint analysis and that you construct the test statistic using the beta parameter as was done in the summer 2015 analysis. Then you can reference the tech note for the summer 2016 analysis.
778-806: I think you need to introduce the beta parameter here since it is used to define the null hypothesis. It is also a bit hard to follow the procedure with only text. Can you add equations?
Section 7.3: It is important to make clear in the text when beta is being set for the toy experiments and

when beta is being set to calculate the test statistic. The explanation of the procedure seems to get a little bit lost in the discussion of left and right tails. I think you can start with a simple explanation like: "The beta parameter is defined such that numu-bar -> nue-bar oscillations are non-existant when beta=0 and consistent with the PMNS prediction when beta=1. The null hypotheses for numu-bar -> nue-bar oscillations is defined by setting beta=0. N toys are generated for the null hypothesis (beta=0) and for each toy, the test statistic is calculated. The test statistic is defined as chi^2(beta=0)-chi^2(beta=1). The distribution of the test statistic from the toy experiments is saved. The test statistic is also calculated for the data. The p-value is defined as the fraction of toy experiments with a test statistic value that is larger than the value from the data."

On Apr 8, 2016, at 8:37 AM, Claudio Giganti <cgiganti@lpnhe.in2p3.fr> wrote:

Hi Davide and Raj,

thanks for releasing this version of the TN. I went through it and I think it's in really good shape (and the results are very nice!)

I see no-problems in releasing it to the whole OA group and once you get some comments also from Mark to release it to the reviewers at the beginning of next week.

My main comment is about the content of this preliminary version when you will send it to the reviewers.
Do we want to have the nue-bar and numu-bar sections already included?
I don't have a strong opinion but it might be better to send to the reviewers only the joint part and then add the nue-bar and numu-bar sections once they are completed.

Below are some comments on the version you sent to us.

Cheers,
Claudio

Since you mention from the begin Tab. 12 (even in the abstract) it might be better to move it a bit earlier.
L. 198: 4.0 and not 0.4 POT Done
L. 309 should be mu-like and 312 e-like It should be correct That bin contains almost 0 entries.
Fig.5: why there is one event at (0,0) in the e-like RHC plot? There is not cut for low reco energy
L. 435: If I understand correctly the MEC nu-bar component is given by the product of the MEC nu * MEC nubar. If that's the case it should be 1.546*0.578=0.89. Is that correct? Done
If that's the case I think you should write it in the text since it's a bit misleading to compare 0.578 to 1.031 from the previous BANFF fit.
L. 468: broken reference to Tab. Done
Tab. 13, 15 and 16: I would remove the first line with all syst Done
Tab. 17, 18, 19, 20: can we add a line with the Flux+Xsec for parameters constrained by ND280? Done
Fig. 20: when we show the 2D th13-dcp plot with T2K only it might be nice to put also the area corresponding to 1-sigma reactor constraint (especially in Fig.28 with the fit on the data) Done
Sect. 5.3: are we going to include this discussion on the pulls? As we decided a few months ago, coverage should be enough
L. 734: 3 is also in perfect agreement with the Asimov B with dCP=0 ;-) Done

On 07/04/2016, at 00:17, Davide Sgalaberna <davide.sgalaberna@cern.ch> wrote:

Hi,

Raj and I have prepared a preliminary version of the VALOR TN, that you can find attached.
Most of the things are there: event rate, systematic tables and plots, Asimov fits as well as Run 1-6 data fits.
Some things are still missing but most is done.
The nuebar appearance and numubar disappearance analyses are already included.
Rate-only nuebar p-values are still missing and will be run in the next days.
For what concern numubar disappearance we already put some Asimov contours.

Please let us know if you have comments and suggestions so we can address them before sending to the OA mailing list.

What would you like we present tomorrow? The data fit is ready and we can present it.
However if you prefer to wait that all the OA groups have fitted the data we can postpone it to the next OA meeting.
We are also going to show and update about NueBar and NumuBar sensitivities.

Best regards,
Davide

<jointfit2016.pdf>

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